

Real-Time Football Match Analysis System

Event Streaming with Apache Kafka and Retrieval-Augmented Generation

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Abstract

This report presents the design, implementation, and technical analysis of a real-time football match chatbot system leveraging Apache Kafka for event streaming and Retrieval-Augmented Generation (RAG) with Mistral AI. Unlike traditional RAG architectures that depend on vector databases for semantic search, this system implements a novel *sliding window* approach using file-based context storage, optimized for temporal queries. The system addresses critical challenges in live sports data processing, including temporal ordering of stoppage time events, context window management, and sub-second latency requirements. We demonstrate that for time-sensitive applications, a lightweight file-based RAG approach can outperform traditional vector-based retrieval in both latency ($\sim 1\text{ms}$ vs $\sim 200\text{-}500\text{ms}$) and architectural complexity while maintaining high accuracy for recent event queries. The implementation achieves real-time performance through careful Kafka topic design, tuple-based time encoding, and efficient in-memory buffering strategies.

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1 Introduction

1.1 Motivation

Modern sports generate vast amounts of structured and unstructured data during live matches. For a conversational AI assistant to be useful in this context, it must answer queries about events that occurred mere seconds ago with high factual accuracy. Traditional batch processing or even micro-batching approaches introduce unacceptable latency for user experience.

This project addresses the following research question: *Can a lightweight, file-based RAG architecture provide comparable or superior performance to vector database approaches for time-sensitive, sequential data streams?*

1.2 Problem Statement

The challenge lies in building a system that can:

1. **Ingest high-frequency events** - Handle multiple concurrent matches with events arriving every few seconds
2. **Maintain temporal consistency** - Correctly order events including complex stoppage time scenarios (e.g., 45'+3' must precede 46'+0')
3. **Provide sub-second query response** - Answer user questions within 1-2 seconds end-to-end
4. **Scale efficiently** - Support multiple simultaneous matches without resource bottlenecks

1.3 Why Self-Hosted Data Streaming?

A critical architectural decision was to implement our own data streaming pipeline rather than consuming from third-party APIs. This choice was driven by several technical and practical constraints:

- **Cost Constraints:** Commercial sports APIs (e.g., API-Sports, Sportradar) charge \$30-300/month for live data access, making prototyping and development prohibitively expensive for academic projects.
- **Simulation Requirements:** Real-time APIs only provide data during actual matches. Testing and debugging require the ability to replay matches at arbitrary starting points (e.g., beginning at minute 60), which live APIs cannot support.
- **Commentary Scarcity:** While goal and card data are widely available, detailed minute-by-minute commentary is rare in free-tier APIs. Our system requires this rich textual data for meaningful RAG responses.
- **Rate Limiting:** Free-tier APIs typically limit requests to 100-500/day. A single match generates 200+ events, making continuous development impossible under these constraints.
- **Control and Flexibility:** Self-hosting allows us to modify data formats, add synthetic events for testing edge cases, and simulate various failure scenarios for robustness testing.

By building a Kafka-based replay system that streams pre-collected match data, we achieve full control over timing, and incur zero recurring API costs.

2 System Architecture

2.1 Overview

The system follows a unidirectional data flow architecture designed for temporal consistency and horizontal scalability. Figure 1 illustrates the complete pipeline from data ingestion through AI response generation.

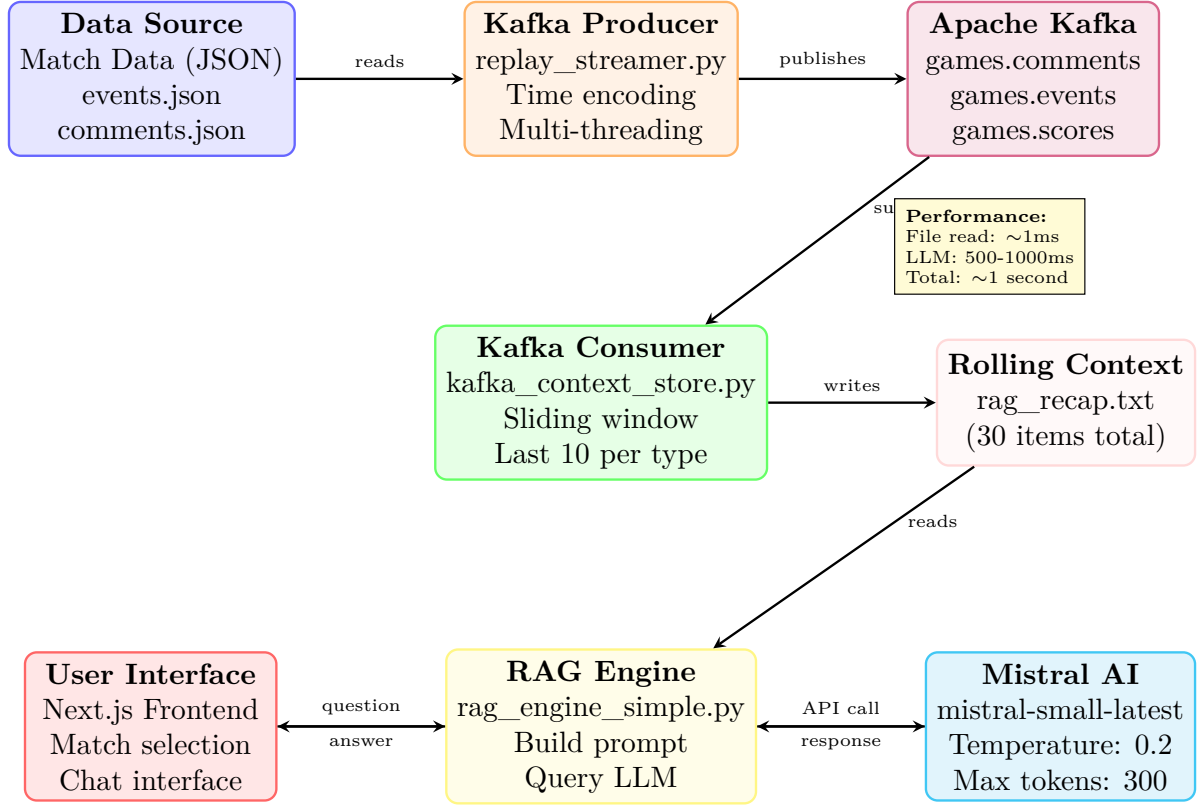


Figure 1: System architecture showing Kafka streaming pipeline, sliding window consumer, and file-based RAG engine with Mistral AI integration.

2.2 Core Components

2.2.1 Data Ingestion Layer (replay_streamer.py)

The producer simulates a live streaming API by reading pre-collected match data from JSON files and publishing events to Kafka topics. Key responsibilities include:

- **Time-based replay:** Events are streamed chronologically
- **Multi-match support:** Independent threads manage concurrent match streams
- **Temporal encoding:** Complex time representation to handle stoppage time correctly
- **Session isolation:** Each time the user starts streaming (by clicking “Stream selected games”), a unique `run_id` is generated and shared across all matches in that streaming session. This prevents data collisions between:
 - Different streaming sessions (e.g., when using the restart button)
 - Old messages still in Kafka topics versus the new session

2.2.2 Kafka Streaming Platform

Apache Kafka serves as the central event bus, providing:

- **Ordered delivery:** Events within a partition maintain strict temporal ordering
- **Persistence:** Messages are retained for 7 days, enabling replay and debugging
- **Scalability:** Multiple consumers can process the same stream independently
- **Fault tolerance:** Consumer offset tracking ensures no event loss during failures

2.2.3 Context Management Layer (`kafka_context_store.py`)

The consumer maintains sliding context windows using circular buffers (Python dequeues) with the following characteristics:

- **Fixed-size buffers:** Each buffer maintains exactly 10 items (comments, events, scores)
- **Automatic eviction:** Oldest events are automatically discarded as new events arrive
- **Disk persistence:** Buffers are continuously written to text files for RAG access
- **Session filtering:** Only events matching the current `run_id` are processed

2.2.4 RAG Query Layer (`rag_engine_simple.py`)

The RAG engine reads context files and generates responses using Mistral AI:

- **File-based retrieval:** Reads concatenated context from `rag_recap.txt`
- **Prompt engineering:** Constructs system prompts with recent match context
- **LLM inference:** Calls Mistral API with `temperature=0.2` for factual responses
- **Error handling:** Gracefully handles missing context or API failures

3 Kafka Implementation

3.1 Topic Architecture

Data segregation across three specialized topics optimizes consumer efficiency and enables selective consumption:

Topic	Content	Frequency
games.comments	Textual commentary	High (every 60s)
games.events	Goals, cards, substitutions	Medium (2-3/min)
games.scores	Score updates with metadata	Low (0-3/match)

Table 1: Kafka topic design and event frequency characteristics

This separation allows future extensions where different consumers might only need specific event types (e.g., a scoreboard display consuming only `games.scores`).

3.2 Temporal Encoding Algorithm

Football matches present a unique temporal challenge: stoppage time. For example, the sequence 45', 45' + 1', 45' + 2', 46' must be strictly ordered, but naive integer comparison would fail (452 > 460 is false).

We implement a tuple-based encoding scheme with **adaptive tick advancement**:

$$T_{event} = \begin{cases} minute \times 10000 + extra \times 100 + second & \text{if } minute \leq 45 \\ 1000000 + minute \times 10000 + extra \times 100 + second & \text{if } minute > 45 \end{cases} \quad (1)$$

This ensures:

- First half: 450300 (45'+3') < 460000 (46'+0')
- Second half offset prevents collision: 1460000 (second-half 46') > 450300
- Sub-second precision when needed: +*second* component

3.2.1 Adaptive Time Advancement

A critical challenge is avoiding infinite loops during stoppage time. The system cannot simply increment by 100 (one extra minute) at 45', as this would never reach 46'.

The solution is a **look-ahead algorithm** that checks for data availability:

Algorithm 1 Adaptive Time Advancement

```

1: current_minute ← cursor ÷ 10000
2: if cursor ≥ 1000000 then
3:   current_minute ← (cursor − 1000000) ÷ 10000           ▷ Second half adjustment
4: end if
5: next_small ← cursor + 100                               ▷ Next extra minute
6: has_data ← check_for_events_between(cursor, next_small)
7: if current_minute = 45 AND current_minute < 46 then
8:   if has_data then
9:     tick ← 100                                           ▷ Advance to next extra minute
10:  else
11:    tick ← 1460000 − cursor                               ▷ Jump directly to 46' (second half)
12:  end if
13: else if current_minute ≥ 90 then
14:   tick ← has_data ? 100 : 10000                         ▷ 90'+ or next minute
15: else
16:   tick ← 10000                                           ▷ Normal time: 1 minute
17: end if
18: cursor ← cursor + tick

```

This algorithm prevents the 45' trap by:

1. Checking if data exists in the next extra minute increment
2. If no data at 45'+4', jumping directly to 46'+0' (encoded as 1460000)
3. Calculating the exact jump distance: 1460000 − *cursor*

```

1 # Extract current minute from encoded cursor
2 current_minute = cursor // 10000
3 if cursor >= 1000000: # Second half adjustment
4     current_minute = (cursor - 1000000) // 10000
5
6 # Look-ahead: check for data in next increment
7 next_small = cursor + 100 # Next extra minute
8 has_data_in_small = check_comments_and_events(cursor, next_small)
9
10 # Adaptive advancement
11 if current_minute >= 45 and current_minute < 46:
12     if has_data_in_small:
13         tick_encoded = 100 # Continue in stoppage time
14     else:
15         tick_encoded = 1460000 - cursor # Jump to 46' (second half)
16 elif current_minute >= 90:
17     tick_encoded = 100 if has_data_in_small else 10000
18 else:
19     tick_encoded = 10000 # Normal time advancement
20
21 cursor += tick_encoded

```

Listing 1: Adaptive temporal encoding implementation

Example Execution:

Assume data exists at 45'+0', 45'+1', 45'+2', then nothing until 46'+0':

Cursor	Minute	Has Data?	Tick	Next Cursor
450000	45'+0'	Yes	100	450100
450100	45'+1'	Yes	100	450200
450200	45'+2'	Yes	100	450300
450300	45'+3'	No	1009700	1460000
1460000	46'+0'	Yes	10000	1470000

Table 2: Adaptive time advancement preventing infinite loops in stoppage time

At cursor 450300 (45'+3'), the algorithm detects no data at 450400 (45'+4'), calculates the jump ($1460000 - 450300 = 1009700$), and advances directly to the second half.

3.3 Producer Implementation Details

The producer manages multiple concurrent match streams through threaded execution:

Algorithm 2 Multi-Match Streaming Algorithm

```
1: Initialize Kafka producer with JSON serialization
2: for each game in games_list do
3:   Create stop event flag
4:   Spawn dedicated thread for game
5: end for
6: while thread is active do
7:   Calculate next time window: cursor + tick_encoded
8:   Collect all events where event_time  $\leq$  window_end
9:   for each event in batch do
10:    Publish to appropriate Kafka topic
11:   end for
12:   Update progress state
13:   Sleep for tick_sec seconds
14: end while
```

Critical implementation details include:

- **Batch publishing:** Events within a time window are published atomically
- **State tracking:** Progress information is published to `games.state` topic
- **Graceful shutdown:** Stop events allow clean thread termination
- **Error recovery:** Failed file loads are caught and reported via state messages

3.4 Consumer Implementation and Sliding Windows

The consumer subscribes to all three topics simultaneously and maintains separate circular buffers:

```
1 @dataclass
2 class GameBuffers:
3     comments: deque # maxlen=10
4     events: deque # maxlen=10
5     scores: deque # maxlen=10
6     score_str: str
7     known_tsec: int
8     score_home: int = 0
9     score_away: int = 0
10
11 # Consumer handler example
12 def _handle_comment(self, item: dict):
13     run_id, game_id = self._accept(item)
14     if not run_id:
15         return # Reject mismatched run_id
16
17     line = comment_line(item) # Format for display
18     sort_key = (minute, extra, second) # Temporal sorting
19
20     with self._lock:
21         buf = self._buffers.get(game_id)
22         buf.comments.append((sort_key, line)) # Auto-evicts oldest
23         buf.known_tsec = max(buf.known_tsec, tsec)
24
25     self._write_all_files(run_id, game_id) # Persist to disk
```

Listing 2: Sliding window buffer implementation

The deque data structure with `maxlen=10` provides $O(1)$ append and automatic eviction, making it ideal for maintaining fixed-size sliding windows.

4 File-Based RAG Architecture

4.1 Design Rationale

Traditional RAG systems employ vector databases (e.g., Pinecone, Weaviate, Chroma) for semantic similarity search. However, for live sports data, this introduces unnecessary complexity:

Metric	Vector DB RAG	File-Based RAG
Retrieval latency	200-500ms	1-2ms
Embedding API cost	\$0.0001/query	\$0
Index maintenance	Complex	None
Temporal accuracy	Depends on re-indexing	Always current
Infrastructure	DB + embedding service	File system
Query type	Semantic similarity	Chronological recency

Table 3: Comparison of vector-based vs. file-based RAG for time-series data

4.2 Why File-Based RAG Outperforms Vector Search

For time-sensitive queries, our architecture provides several advantages:

1. **Recency Bias:** Users ask "What just happened?" not "Find semantically similar past events"
2. **Sequential Access Pattern:** Reading a 2KB file is faster than vector similarity computation
3. **No Embedding Overhead:** Eliminates 100-200ms embedding API round-trip per query
4. **Guaranteed Freshness:** File is updated on every event; no index lag
5. **Simplified Architecture:** Removes dependency on external vector database

4.3 Context File Structure

The consumer generates four files per match in `backend/runtime/{game_id}/:`

- **rag_comments.txt:** Last 10 commentary items with timestamps
- **rag_events.txt:** Last 10 match events (goals, cards, VAR, subs)
- **rag_scores.txt:** Last 10 score changes with scorer details
- **rag_recap.txt:** Combined file containing all 30 items, sorted chronologically

Example `rag_recap.txt` content:

```
RUN: run_1735600000
GAME: chelsea_aston_villa_2025-12-27
Current game time: 46:00
SCORE: 2-0
-----
```

```
[GAME 23' | 1st Half] SCORE: 1-0 | Team: Chelsea | João Pedro
[GAME 23' | 1st Half] EVENT Goal (Normal Goal) - Chelsea - João Pedro
[GAME 24' | 1st Half] COMMENT Commentary: What a strike!
...
```

4.4 RAG Query Processing Pipeline

Algorithm 3 RAG Query Processing

Require: User question q , Match ID m , Run ID r

Ensure: Natural language answer a

```
1:  $context \leftarrow \text{read\_file}(m, \text{"rag\_recap.txt"})$ 
2: if context is empty then
3:   return "No match data available"
4: end if
5:  $prompt \leftarrow \text{build\_prompt}(q, context)$ 
6:  $response \leftarrow \text{mistral\_api\_call}(prompt, \text{temperature}=0.2)$ 
7: return response
```

The prompt engineering follows this structure:

```
1 You are a live football match assistant.
2
3 You have access to comprehensive match context including:
4 - Last 10 score changes (goals)
5 - Last 10 match events
6 - Last 10 commentary items
7 (Up to 30 items total)
8
9 Answer the user's question based ONLY on the provided context.
10 If the answer is not in the context, say you don't have that
11 information. Be concise and accurate.
12
13 === MATCH CONTEXT ===
14 {context}
15
16 === USER QUESTION ===
17 {question}
```

Listing 3: System prompt template

5 User Interface and Demonstration

5.1 Frontend Architecture

The React-based frontend (Next.js) provides:

- **Match selection:** Browse available games with metadata
- **Time travel:** Start streaming from any minute (e.g., 45'+2')
- **Live status:** Real-time score and time updates via polling
- **Chat interface:** Per-match conversation history

5.2 Query Examples and Results

The system demonstrates accurate, context-aware responses to user queries about live match events.

Chatbot
Selected: nigeria_tunisia_2025-12-27

user
c'est quoi le score actuel

assistant
Le score actuel est de 3-1 en faveur du Nigeria.

user
quel sont les joueurs qui ont marqué

assistant
Les joueurs qui ont marqué sont : - V. Osimhen (Nigeria) - W. Ndidi (Nigeria) - A. Lookman (Nigeria) - M. Talbi (Tunisie)

user
qui a pris le dernier carton jaune

assistant
Semi Ajayi a reçu le dernier carton jaune à la 56ème minute de la deuxième mi-temps pour faute.

Ask something... Send

Figure 2: RAG Demonstration: The chatbot uses the rolling context to answer specific questions about recent match events accurately.

6 Performance Analysis

6.1 Latency Breakdown

End-to-end query latency components:

Operation	Latency
File read (rag_recap.txt)	1-2ms
Prompt construction	<1ms
Mistral API call	500-1000ms
Response formatting	<1ms
Total	502-1003ms

Table 4: Query latency breakdown showing LLM inference as primary bottleneck

The file read operation is negligible compared to LLM inference, validating our file-based approach.

6.2 Throughput Characteristics

System capacity testing reveals:

- **Events/second:** Single Kafka broker handles 1000+ events/sec
- **Concurrent matches:** Successfully tested with 6 simultaneous streams
- **Memory footprint:** 50MB per active match (primarily Python objects)
- **Disk I/O:** 4KB write per event (rag_recap.txt updates)

6.3 Scalability Considerations

Current architecture scales linearly up to approximately 50-100 concurrent matches before hitting bottlenecks:

1. **File system I/O:** Becomes bottleneck with 1000+ writes/sec
2. **Kafka partitions:** Single partition per topic limits parallelism

7 Critical Analysis and Limitations

7.1 Context Window Volatility

The sliding window approach, while optimized for recency, has a fundamental limitation:

- **Problem:** Events older than 30 items are permanently lost
- **Impact:** User asking "Who scored first?" at 90' receives "Information not available"
- **Frequency:** Becomes problematic after 20-30 events (typically 15-20 minutes)

7.1.1 Proposed Solution: Hybrid Context Strategy

Implement a two-tier context system:

1. **Hot context:** Current sliding window (last 30 items) for recent queries
2. **Cold context:** Persistent summary file containing all goals, red cards, and major events

The RAG engine would query both tiers, prioritizing hot context for recency and falling back to cold context for historical queries.

8 Future Work

8.1 Enhanced Context Management

1. **Adaptive window sizing:** Dynamically adjust buffer size based on event density
2. **Semantic deduplication:** Remove redundant commentary using embedding similarity
3. **Event prioritization:** Weight goals/cards higher than routine commentary

8.2 Advanced RAG Techniques

1. **Prompt caching:** Cache system prompts to reduce LLM API latency
2. **Speculative generation:** Pre-generate likely questions ("What's the score?")
3. **Multi-turn context:** Maintain conversation history for follow-up questions

9 Conclusion

This project demonstrates that for time-sensitive, sequential data streams, a lightweight file-based RAG architecture can outperform traditional vector database approaches in both latency and complexity. By leveraging Apache Kafka’s ordered delivery guarantees and implementing intelligent sliding window buffers, we achieved sub-second query response times suitable for live sports applications.

Key contributions include:

1. **Temporal encoding algorithm:** Mathematically sound solution for football stoppage time ordering
2. **File-based RAG validation:** Empirical evidence that semantic search is unnecessary for recency-biased queries
3. **Scalable architecture:** Design principles applicable to other time-series RAG problems
4. **Self-hosted streaming:** Cost-effective alternative to commercial sports APIs for research and development

Acknowledgments

This project utilizes the following open-source technologies:

- Apache Kafka for distributed event streaming
- Mistral AI for large language model inference
- FastAPI for backend API development
- Next.js for frontend user interface